

# ST. XAVIER'S COLLEGE

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LAB/WRITTEN ASSIGNMENT NUMBER: 4.2

“MS-Excel”

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## Cell in Microsoft Excel

A cell is the basic unit of a worksheet in Microsoft Excel, formed by the intersection of a row and a column. It is used to enter, store, and manipulate data such as text, numbers, and formulas.

Example: The intersection of column B and row 3 is called cell B3.

## Relative Cell Reference

A relative cell reference is a cell reference that changes automatically when a formula is copied or moved to another cell.

Example:

If  $=A1+B1$  is written in cell C1 and copied to C2, it changes to  $=A2+B2$ .

## Mixed Cell Reference

A mixed cell reference is a reference in which either the row or the column is fixed, while the other can change.

Example:

$\$A1$  — Column A is fixed, but the row can change.

Or  $A\$1$  — Row 1 is fixed, but the column can change.

## Absolute Cell Reference

An absolute cell reference is a cell reference in which both the row and column are fixed, so the reference does not change when the formula is copied.

Example:

$\$A\$1$  — Both column A and row 1 are fixed.

## Interface of MS-Excel

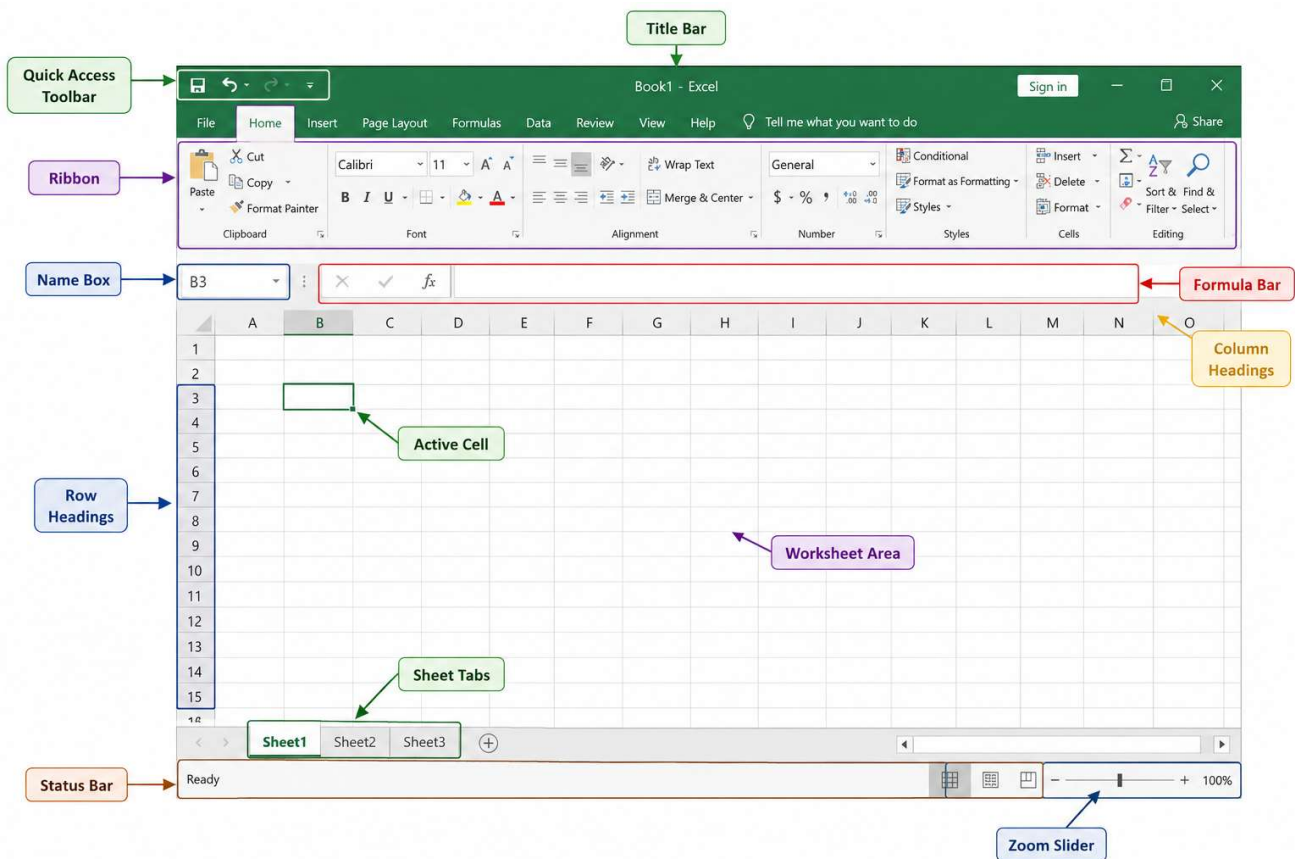


Figure 1 Interface of Excel

## VLOOKUP

VLOOKUP (Vertical Lookup) is an Excel function used to search for a value in the first column of a table and return a corresponding value from another column in the same row.

### Syntax of VLOOKUP

`=VLOOKUP(parameter1, parameter2, argument1, argument2)`

#### Arguments:

lookup\_value – The value we want to search for.

table\_array – The range of cells containing the data.

col\_index\_num – The column number from which the result is returned.

range\_lookup – Specifies the type of match:

TRUE – Approximate match

FALSE – Exact match

|   | A        | B     | C     |
|---|----------|-------|-------|
| 1 | Roll No. | Name  | Marks |
| 2 | 101      | Ram   | 85    |
| 3 | 102      | Shyam | 78    |
| 4 | 103      | Hari  | 92    |

Figure 2 VLOOKUP Example

For example,

In the given Table, To find the marks of Roll No. 102,

We use the given syntax:

=VLOOKUP(102, A2:C4, 3, FALSE)

Result: 78

Here, 102 is the lookup value, A2:C4 is the table range, 3 means the third column (Marks), and FALSE specifies an exact match.

## Sheet 1: Student Marksheet Calculation:

Formulas Used:

### 1)Total:

```
=D3+E3+F3+G3+H3+I3+J3+K3+L3+M3+N3+O3
```

Figure 3Total Formula

### 2)Percentage:

```
=P3/6
```

Figure 4Percentage Formula

### 3)Remarks:

```
=IF(OR(D3<27,E3<8,F3<27,G3<8,H3<27,I3<8,J3<27,K3<8,L3<27,M3<8,N3<17,O3<17),"FAIL","PASS")
```

Figure 5Remarks Formula

### 4)Grade

```
=IF(Q3="FAIL","NG",IF(R3>90,"A+",IF(R3>80,"B",IF(R3>70,"B+",IF(R3>60,"B",IF(R3>50,"C+",IF(R3>40,"D"))))))
```

Figure 6Grade Formula

### 5)Rank:

```
=IF(Q4="FAIL","-",RANK(R4,$R$3:$R$12))
```

Figure 7Rank Formula

## ST. XAVIER'S COLLEGE

SREYAM THAPA Roll No:3

| ROLL | NAME              | SECTION | PHYSICS | PHYSICS | CHEMISTRY | CHEMISTRY | MATHS | MATHS | PHENOLISH | ENGLISH | PAINEPAL | THE NEPAL | COMPUTER | COMPUTER | PA  | TOTAL | REMARK    | PERCENTAGE | GRADE |
|------|-------------------|---------|---------|---------|-----------|-----------|-------|-------|-----------|---------|----------|-----------|----------|----------|-----|-------|-----------|------------|-------|
| 1    | Hai Shresha       | 1       | 64      | 23      | 45        | 23        | 54    | 25    | 12        | 25      | 66       | 21        | 45       | 45       | 448 | FAIL  | 74.666667 | NG         |       |
| 2    | Shyam Thapa       | 1       | 45      | 21      | 54        | 21        | 56    | 24    | 34        | 24      | 54       | 21        | 44       | 45       | 443 | PASS  | 73.833333 | B+         |       |
| 3    | Sanskar Bakhel    | 1       | 32      | 24      | 63        | 24        | 65    | 25    | 63        | 23      | 67       | 24        | 43       | 45       | 498 | PASS  | 83        | A          |       |
| 4    | Budesh Dhungana   | 1       | 66      | 25      | 68        | 23        | 45    | 24    | 35        | 22      | 66       | 23        | 42       | 45       | 484 | PASS  | 80.666667 | A          |       |
| 5    | Cristiano Ronaldo | 1       | 47      | 21      | 56        | 21        | 43    | 23    | 65        | 21      | 54       | 21        | 45       | 45       | 462 | PASS  | 77        | B+         |       |
| 6    | Lionel Messi      | 1       | 72      | 21      | 74        | 21        | 75    | 23    | 61        | 25      | 68       | 25        | 45       | 45       | 555 | PASS  | 92.5      | A+         |       |
| 7    | Nico Williams     | 1       | 45      | 23      | 34        | 23        | 32    | 22    | 43        | 24      | 43       | 23        | 43       | 45       | 400 | PASS  | 66.666667 | B          |       |
| 8    | Pedri Gonzalez    | 1       | 65      | 14      | 56        | 24        | 32    | 24    | 66        | 23      | 33       | 21        | 32       | 45       | 435 | PASS  | 72.5      | B+         |       |
| 9    | Rodriggo          | 1       | 57      | 21      | 54        | 16        | 67    | 25    | 44        | 22      | 32       | 22        | 33       | 45       | 438 | PASS  | 73        | B+         |       |
| 10   | Alex Yadav        | 1       | 33      | 16      | 33        | 19        | 66    | 22    | 21        | 21      | 54       | 21        | 34       | 41       | 381 | FAIL  | 63.5      | NG         |       |

Figure 8 Student Marksheets

## Sheet 2: VLOOKUP Based Student Result:

A sample of a VLOOKUP based student result is given in the below figure:

| ST. XAVIER'S COLLEGE |               |            |                |
|----------------------|---------------|------------|----------------|
| Name                 | Hari Shrestha |            |                |
| Roll No.             | 1             |            |                |
| Section              | I             |            |                |
| Marksheet            |               |            |                |
| Subjects             | Full Marks    | Pass Marks | Obtained Marks |
| Physics TH           | 75            | 27         | 64             |
| Physics PR           | 25            | 8          | 23             |
| Chemistry TH         | 75            | 27         | 45             |
| Chemistry PR         | 25            | 8          | 23             |
| MATHS TH             | 75            | 27         | 54             |
| MATHS PR             | 25            | 8          | 25             |
| ENGLISH TH           | 75            | 27         | 12             |
| ENGLISH PR           | 25            | 8          | 25             |
| NEPALI TH            | 75            | 27         | 66             |
| NEPALI PR            | 25            | 8          | 21             |
| COMPUTER TH          | 50            | 17         | 45             |
| COMPUTER PR          | 50            | 17         | 45             |
| RESULT:              | FAIL          |            |                |
| GRADE:               | NG            |            |                |
| RANK:                | -             |            |                |

Figure 9 VLOOKUP Based Student Result

## VLOOKUP BASED RESULTS OF 5 STUDENTS:

| ST. XAVIER'S COLLEGE |               |            |                |
|----------------------|---------------|------------|----------------|
| Name                 | Hari Shrestha |            |                |
| Roll No.             | 1             |            |                |
| Section              | I             |            |                |
| Marksheet            |               |            |                |
| Subjects             | Full Marks    | Pass Marks | Obtained Marks |
| Physics TH           | 75            | 27         | 64             |
| Physics PR           | 25            | 8          | 23             |
| Chemistry TH         | 75            | 27         | 45             |
| Chemistry PR         | 25            | 8          | 23             |
| MATHS TH             | 75            | 27         | 54             |
| MATHS PR             | 25            | 8          | 25             |
| ENGLISH TH           | 75            | 27         | 12             |
| ENGLISH PR           | 25            | 8          | 25             |
| NEPALI TH            | 75            | 27         | 66             |
| NEPALI PR            | 25            | 8          | 21             |
| COMPUTER TH          | 50            | 17         | 45             |
| COMPUTER PR          | 50            | 17         | 45             |
| RESULT:              | FAIL          |            |                |
| GRADE:               | NG            |            |                |
| RANK:                | -             |            |                |

Figure 11Result-1

| ST. XAVIER'S COLLEGE |             |            |                |
|----------------------|-------------|------------|----------------|
| Name                 | Shyam Thapa |            |                |
| Roll No.             | 2           |            |                |
| Section              | I           |            |                |
| Marksheet            |             |            |                |
| Subjects             | Full Marks  | Pass Marks | Obtained Marks |
| Physics TH           | 75          | 27         | 45             |
| Physics PR           | 25          | 8          | 21             |
| Chemistry TH         | 75          | 27         | 54             |
| Chemistry PR         | 25          | 8          | 21             |
| MATHS TH             | 75          | 27         | 56             |
| MATHS PR             | 25          | 8          | 24             |
| ENGLISH TH           | 75          | 27         | 34             |
| ENGLISH PR           | 25          | 8          | 24             |
| NEPALI TH            | 75          | 27         | 54             |
| NEPALI PR            | 25          | 8          | 21             |
| COMPUTER T           | 50          | 17         | 44             |
| COMPUTER F           | 50          | 17         | 45             |
| RESULT:              | PASS        |            |                |
| GRADE:               | B+          |            |                |
| RANK:                | 6           |            |                |

Figure 10Result2

| ST. XAVIER'S COLLEGE |            |           |                |
|----------------------|------------|-----------|----------------|
| Name                 | Alex Yadav |           |                |
| Roll No.             | 10         |           |                |
| Section              | I          |           |                |
| Marksheet            |            |           |                |
| Subjects             | Full Marks | Pass Mark | Obtained Marks |
| Physics TH           | 75         | 27        | 33             |
| Physics PR           | 25         | 8         | 16             |
| Chemistry TH         | 75         | 27        | 33             |
| Chemistry PR         | 25         | 8         | 19             |
| MATHS TH             | 75         | 27        | 66             |
| MATHS PR             | 25         | 8         | 22             |
| ENGLISH TH           | 75         | 27        | 21             |
| ENGLISH PR           | 25         | 8         | 21             |
| NEPALI TH            | 75         | 27        | 54             |
| NEPALI PR            | 25         | 8         | 21             |
| COMPUTER T           | 50         | 17        | 34             |
| COMPUTER F           | 50         | 17        | 41             |
| RESULT:              | FAIL       |           |                |
| GRADE:               | NG         |           |                |
| RANK:                | -          |           |                |

Figure 12Result3

| ST. XAVIER'S COLLEGE |              |           |                |
|----------------------|--------------|-----------|----------------|
| Name                 | Lionel Messi |           |                |
| Roll No.             | 6            |           |                |
| Section              | I            |           |                |
| Marksheet            |              |           |                |
| Subjects             | Full Marks   | Pass Mark | Obtained Marks |
| Physics TH           | 75           | 27        | 72             |
| Physics PR           | 25           | 8         | 21             |
| Chemistry TH         | 75           | 27        | 74             |
| Chemistry PR         | 25           | 8         | 21             |
| MATHS TH             | 75           | 27        | 75             |
| MATHS PR             | 25           | 8         | 23             |
| ENGLISH TH           | 75           | 27        | 61             |
| ENGLISH PR           | 25           | 8         | 25             |
| NEPALI TH            | 75           | 27        | 68             |
| NEPALI PR            | 25           | 8         | 25             |
| COMPUTER T           | 50           | 17        | 45             |
| COMPUTER F           | 50           | 17        | 45             |
| RESULT:              | PASS         |           |                |
| GRADE:               | A+           |           |                |
| RANK:                | 1            |           |                |

Figure 13Result4

| ST. XAVIER'S COLLEGE |                   |           |                |
|----------------------|-------------------|-----------|----------------|
| Name                 | Cristiano Ronaldo |           |                |
| Roll No.             | 5                 |           |                |
| Section              | I                 |           |                |
| Marksheet            |                   |           |                |
| Subjects             | Full Marks        | Pass Mark | Obtained Marks |
| Physics TH           | 75                | 27        | 47             |
| Physics PR           | 25                | 8         | 21             |
| Chemistry TH         | 75                | 27        | 56             |
| Chemistry PR         | 25                | 8         | 21             |
| MATHS TH             | 75                | 27        | 43             |
| MATHS PR             | 25                | 8         | 23             |
| ENGLISH TH           | 75                | 27        | 65             |
| ENGLISH PR           | 25                | 8         | 21             |
| NEPALI TH            | 75                | 27        | 54             |
| NEPALI PR            | 25                | 8         | 21             |
| COMPUTER T           | 50                | 17        | 45             |
| COMPUTER F           | 50                | 17        | 45             |
| RESULT:              | PASS              |           |                |
| GRADE:               | B+                |           |                |
| RANK:                | 4                 |           |                |

Figure 14Result5

## Lab Activity Reflections

### **Lab 1: Introduction to Microsoft Excel**

In the first lab, I learned the basics of Microsoft Excel, including the Excel interface, cells, rows, columns, and basic formulas. I also learned how to create a bill and represent data using different types of charts. This lab helped me understand the fundamental features of Excel.

### **Lab 2: Creating a Mark Sheet**

In the second lab, I learned how to create a student mark sheet and perform basic calculations such as total and percentage. I was also introduced to the VLOOKUP function, which helped me understand how data can be searched and retrieved from a table.

### **Lab 3: Complete Student Result**

In the third lab, I learned how to create a complete student mark sheet and result using various Excel formulas. I calculated the total, percentage, grade, remark, and rank, and used VLOOKUP to retrieve student information and marks. Finally, I created a complete individual student result sheet based on the student's roll number.

## Conclusion

The three Excel lab activities helped me develop practical skills in using Microsoft Excel, starting from basic operations and progressing to advanced spreadsheet applications. I learned to create bills and charts, prepare student mark sheets, use various formulas, and apply VLOOKUP to retrieve data. By the end of the labs, I was able to create a complete student result sheet containing total, percentage, grade, remark, and rank. These activities improved my understanding of Excel and its usefulness for organizing, calculating, and presenting data efficiently.